

Project Title: Stock Analysis Dashboard

Project URL: <https://python-dev-intern-projects-bat6h4lnmqikaeuuxpsuwp.streamlit.app/>

Objective: This project aims to facilitate the analysis and visualization of stock trends using key technical indicators. The dashboard is designed to assist investors, traders, and financial analysts in making informed decisions by visualizing stock trends and understanding market momentum.

Tools and Technologies:

This project is Python-based and utilizes the following libraries:

- **yfinance:** To download market data from the Yahoo! Finance API.
- **Plotly:** An open-source interactive data visualization library for Python.
- **Streamlit:** To rapidly create and share custom web applications.

Stock Terminology:

1. Stock Ticker

Stock symbols, also referred to as ticker symbols, are a shorthand way of identifying a company's stock on an exchange. For example:

Individual Stocks: Apple Inc. (AAPL), Alphabet Inc. (GOOGL), Microsoft Corporation (MSFT), Amazon.com, Inc. (AMZN), Meta (formerly Facebook) Inc. (META), Tesla Motors (TSLA), The Goldman Sachs Group, Inc. (GS)

Market Indices: The Dow Jones Industrial Average (DJIA), The S&P 500 Index (SPX), The NASDAQ Composite Index (COMP)

2. Technical Indicators

These indicators provide insights into stock price movements and market trends:

- **Simple Moving Average (SMA):** The SMA calculates the average price of an asset over a specified number of periods.
- **Exponential Moving Average (EMA):** The EMA gives more weight to recent prices, making it more responsive to new information compared to the SMA.
- **Relative Strength Index (RSI):** The RSI is a momentum oscillator that measures the magnitude of recent price changes to evaluate overbought or oversold conditions of an asset's price, typically over 14 periods.

Traders often use these indicators in combination to identify potential buy and sell signals.

Performance Summary:

Strategy Logic: Buy signals occur when the 50-day EMA crosses above the 100-day SMA *and* the 14-period RSI is above 30. Sell signals are generated when the 50-day EMA crosses below the 100-day SMA *or* the RSI exceeds 70

Stock Analysis Dashboard Implementation

1. User Interface (UI) Design:

- The user interface (UI) was built using Streamlit.

- The dashboard includes an input dropdown to accept stock ticker symbols.
- Once a ticker is selected, users have two primary options:
 - **Generate Graph:** Displays an interactive chart of the selected stock.
 - **Export as CSV:** Exports the stock data displays the processed data and a download link is provided, allowing users to save the data as a CSV file for offline analysis

2. Data Acquisition and Processing:

- Both the "Generate Graph" and "Export as CSV" options fetch historical stock data for the past year using the yfinance library.
- yfinance serves as the primary data source, retrieving market data from the Yahoo! Finance API.

3. Technical Indicator Calculation:

- Technical indicators are calculated over different periods to identify trends and patterns.
- The following indicators are calculated: SMA_20, SMA_50, SMA_100, EMA_20, EMA_50, RSI_14, Buy Signal, and Sell Signal.
- The numeric suffix in the indicator names indicates the period duration (e.g., SMA_20 represents the Simple Moving Average over 20 periods).
- The data has been pre-processed to remove any null or missing values.

4. Chart Generation:

- A Plotly figure is generated with three subplots:
 - A candlestick chart displaying price action, along with SMA and EMA overlays.
 - An RSI indicator to assess momentum.
 - Markers indicating buy and sell signals based on the implemented strategy.



This dashboard visualizes stock trends and market momentum using key technical indicators like SMA, EMA, and RSI to aid investment decisions.